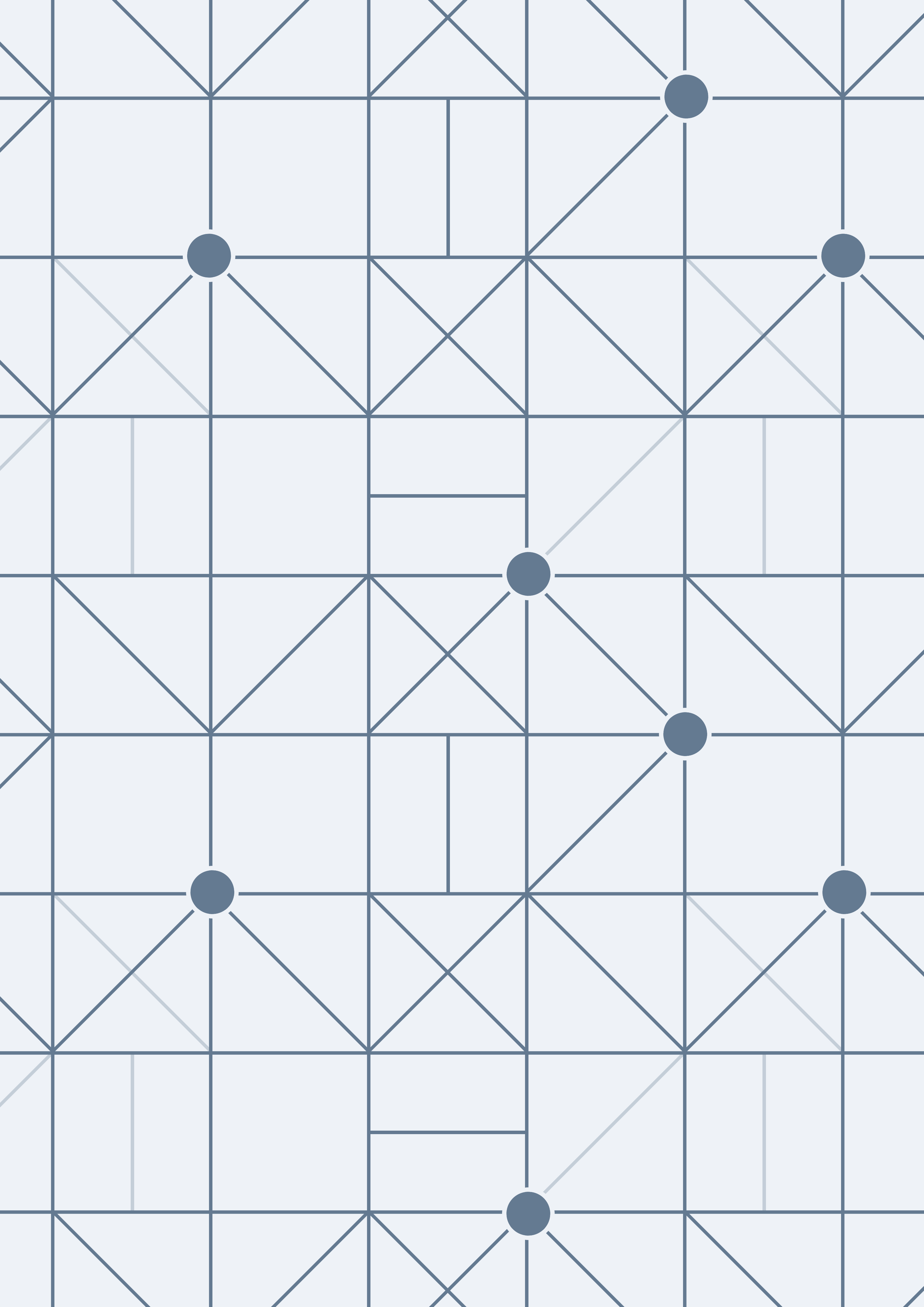

Ideal systems

Ideal system $\Rightarrow$ I can express $H = \sum_i h_i$

In macrocanonical ensemble

$p(C) = \frac{e^{-\beta H(C)}}{Z} = \frac{e^{-\beta \sum_i h_i}}{Z} = \prod_i \frac{e^{-\beta h_i}}{Z}$ $\rightarrow$ In probability if $A \cap B = \emptyset$ then $A \cup B = A \cdot B$, then we obtained result in our $p(C)$.

If I have a potential $V(r_{ij}) \Rightarrow H \neq \sum_i h_i$, but if I have a potential $V(r_i)$ then $H = \sum_i h_i = \sum_i \frac{p_i^2}{2m_i} + \sum_i V(r_i)$

relative position. depends on surrounding particles

only depends on particle i position

for examples in ising model $H = -\frac{J}{2} \sum_{i,j}^{N} S_{i,j} (S_{i+1,j} + S_{i-1,j} + S_{i,j+1} + S_{i,j-1}) \neq \sum_i h_i$

interact with surroundings

$H = -J \sum S_i = -\sum_i J S_i$

Do not interact with surroundings

Boltzmann Gas

It is common limit of fermions and bosons

level of energy of particles

$(l_1, l_2, l_3, l_4, l_5) = (0, 3, 1, 0, 2)$

$(n_0, n_1, n_2, n_3) = (2, 1, 1, 1)$

number of particle in each energy level

Canonical ensemble

$Z = \sum_{\{n_i\}} e^{-\beta \sum n_i \varepsilon_i}$

$\boxed{\sum_i n_i = N}$ $\rightarrow$ this condition can not allow we calculate the partition function as the sum of each partition function

I'm going express Z in terms of $\{l_i\}$

$E = \sum_i n_i \varepsilon_i = \sum_i \varepsilon_{l_i}$

$\Rightarrow Z = \sum_{l_1, \dots, l_N} \boxed{}\, e^{-\beta \sum_i \varepsilon_{l_i}}$

I need express the degeneracy

↳ As particle are indistinguishable, I consider all particle the same, thus we must know how to distribute N particles in l energy levels

$\rightarrow$ A set of $\{n_l\}$ correspond $\frac{N!}{\prod n_l!}$ possible values of $\{l_1, l_2, \dots, l_N\}$ $\longrightarrow$ $Z = \sum_{l_1, \dots, l_N} \frac{\prod n_l!}{N!} e^{-\beta \sum_i^N \varepsilon_{l_i}}$

Boltzmann approximation : $n_l! \simeq 1$

$\rho \lambda_T^3 \ll 1 \Rightarrow$ this approximation is good when $T\uparrow\uparrow$ or $\rho\downarrow\downarrow$

very diluted cases

Under these conditions, bosons and fermions behave the same way

$\lambda_T^2 = \frac{h}{\sqrt{2\pi m K T}}$

With Boltzmann approximation, my partition function is

$Z \simeq \sum_{l_1, \dots, l_N} \frac{1}{N!} e^{-\beta \sum_i \varepsilon_{l_i}} \rightarrow Z \simeq \frac{1}{N!} \left(\sum_{l_1} e^{-\beta \sum_i \varepsilon_{l_i}}\right)\left(\sum_{l_2} e^{-\beta \sum_i \varepsilon_{l_i}}\right) \cdots \left(\sum_{l_N} e^{-\beta \sum_i \varepsilon_{l_i}}\right) = \frac{z^N}{N!}$

$A = -KT \ln Z$ $\qquad$ $A = U - TS$

$S = -KT \sum_C \ln P(C) \cdot P(C)$

$P(C) = \frac{e^{-\beta H(C)}}{Z}$

$S = -K \sum_C \frac{e^{-\beta H(C)}}{Z} \left(-\beta H(C) - \ln(Z)\right) = K\beta \sum_C \frac{e^{-\beta H(C)}}{Z} H(C) + K \ln Z \left(\sum_C \frac{e^{-\beta H(C)}}{Z}\right) Z$ then $S = K\beta U + K \ln Z \left(\sum_C \frac{e^{-\beta H(C)}}{Z}\right) Z$

$p(C)$ $\quad$ energy

$\Downarrow$

$\langle H \rangle = U$ $\rightarrow$ by definition

$A = U - TS \rightarrow S = \frac{U}{T} - \frac{A}{T}$